

MARK LORENZ YAMANAKA

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SUMMARY

Licensed Aeronautical Engineer (PRC, Philippines) with hands-on experience in FEM analysis, ANSYS-based FEA, CAD-based structural design, and validation through engineering calculations. Seeking engineering roles in aerospace, mechanical, and structural engineering, including stress analysis and CAE.

SKILLS

Simulation: ANSYS Workbench, Fluent, Mechanical, SpaceClaim; CalculiX (PrePoMax); OpenFOAM (ParaView)

CAD: SolidWorks, CATIA V6

Programming & Tools: Python, Git/GitHub, XFOIL, Excel, Asana, Power BI, GD&T, OriginLab

Languages: English (Native), Cebuano (Native), Filipino (Professional), Japanese (Conversational)

Certifications: PRC Professional Aeronautical Engineer License (Philippines, 2024)

PROJECTS

SS316 Conical Convergent-Divergent Nozzle | github.com/dmmarkwork-cloud/ss316-cd-nozzle-feasibility

- Ran coupled CFD-FEA (one-way FSI) on an uncooled SS316 C-D nozzle ($P_c = 2$ MPa, $T_c = 800$ K), converging the flow solution to a 0.00003% mass imbalance and confirming $FoS = 7.12$ at the governing junction under combined thermal-pressure load; delivered a GD&T drawing package (ASME Y14.5-2018) with a worst-case tolerance stack closing at $1.68\times$ and a bolted joint ($8\times$ M8 A286) sized to spec.

Aircraft Design Project: MK-10 | github.com/dmmarkwork-cloud/mk10-light-aircraft-design

- Designed a two-seat FAR Part 23 light aircraft (1,649 lb GW) from concept through structure; sized wing loading and power-to-weight via mission/constraint analysis, validated climb, ceiling, and range performance, then built a V-n flight envelope and sized front/rear wing spars from a 10-strip spanwise load distribution (per Bruhn) to a 4.4G utility limit/7.4G ultimate; resolved the design into control surfaces, landing-gear load cases, and a SolidWorks 3-view.

Gas Vessel FEA Analysis | github.com/dmmarkwork-cloud/gas-vessel-fea

- Performed FEM analysis on a thin-walled pressure vessel (3 mm wall, 700 mm length), deriving 3.25 MPa from thin-wall theory and validating against hand calculations ($\leq 2.3\%$ deviation); achieved 0.62% mesh convergence, identified the critical stress location, and evaluated structural safety against design criteria.

EDUCATION

Bachelor of Science in Aeronautical Engineering

Jul 2020 – May 2024

Holy Angel University

- Honors: Dean's List (GPA: 3.6/4.0)
- Relevant Coursework: Aerodynamics, Powerplant, Reliability Engineering, Engineering Data Analysis, Aircraft Systems, Aircraft Structures, Design and System (Aircraft, Helicopter, UAV)

EXPERIENCE

Aircraft Maintenance Engineer Intern | Dornier Technology Inc.

May 2023 – Jul 2023

- Supported 10+ base and line maintenance checks on Airbus A320 / ATR 72-500 / B737-200 / BAE-146 aircraft, ensuring full compliance with CAAP standards and on-time release.
- Participated in 15+ technical inspections (pre and post) to validate safety, helping improve fleet reliability and reduce downtime.
- Assisted in maintenance documentation and audit preparation, contributing to zero non-conformities during internal reviews.

Part-Time Leader / Sub-Leader, Packaging and Logistics | ZOZOBASE

Feb 2018 – May 2026

- Organized and monitored shift activities for a 15-member operations team, maintaining accuracy and schedule adherence during peak demand.
- Identified and suggested workflow improvements that increased throughput by 20% and reduced cycle times by 12%, later adopted by shift supervisors.